Flood Magnitude Prediction Model using Autoregressive and Integrated Moving Average

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ABSTRACT

Disaster is imminent and humans have to learn to manage it. Flood is one of the most common natural disasters in Nigeria with adequate negative impact on people, properties and farmlands. Flooding tends to occur during heavy down pour (rainfall) when water absorption is low. The extent of flood is proportional to the negative impact that is experienced by the victims. The adverse effects of flood cannot be overemphasized. It causes loss of life, properties and economic degradation to mention but a few. These effects can be mitigated via Flood Prediction Models (FPMs). Several successful Flood Prediction Models have been developed, but there are still limitations in the prediction of the magnitude of the flood when it occurs. Flood magnitude is the degree (extent) of the flood. The aim of this study is to build an efficient time series model in forecasting flood magnitude as a natural phenomenon that occurs frequently in nature. In this study, secondary data was collected and used for model training and testing. An autoregressive integrated moving average (ARIMA) Time Series was used to develop a flood magnitude prediction model. The ARIMA model is a combination of autoregressive (AR) and Moving average (MA) models. The implementation was done using Python Spider IDE in ANACONDA for the model simulation. The model was trained, successfully tested and evaluated via coefficient of determination and standard error diagnostic tools. The ARIMA model prediction results indicate that in the year 2024 the flood magnitude will be low level, while that of 2025 will be normal. The result also shows that the year 2026 will have high levels of flood. The results show that the coefficient of AR technique produced a score of 0.0554 indicating a higher and better performance than the MA model with a score value of -0.9077. The higher the coefficient of determination value, the better and more successful the models' performance.

Keywords: Disaster, Flood, Flood Magnitude, Machine Learning, Autoregressive, Moving Average, Time Series.

1. INTRODUCTION

One essential task towards reducing the negative impact of disaster is to adequately and accurately analyze the potential risk and ascertain measures to prevent, mitigate or prepare for emergencies. Information and Communication Technology can play a vital role in highlighting risk areas, vulnerabilities and the likely populations that will be affected by producing geographically referenced analysis such as the use of geographic information system (GIS). Accurate analysis and timely information are keys in mitigating negative impacts of disaster such as flood. In Nigeria several cases of flood were reported by individuals and agencies of government in several towns and communities in 2019 and 2020. Cases of submerged communities and farmlands with negative impacts were common. The negative impact of flood on people and properties can be reduced significantly by improving the flood forecasting system and accurately predicting the extent or magnitude of the flood.

Several Flood Prediction Models (FPM) have been developed and used for this purpose. Different approaches have been applied in developing models to predict flood. These include approaches based on physical parameters (deterministic hydrological models), statistical approaches (statistical models) such as Markov method [1], Machine learning approaches such as Artificial Neural Network[2], Fussy Logic, Time Series and Support Vector Machines [3]. This study is an attempt to apply autoregressive integrated moving average (ARIMA) Time Series algorithm with Machine Learning in the prediction of flood and its magnitude.

The time series is a little bit statistical in nature but its forecasting methods are evaluated using machine learning-like type of techniques [4]. It uses test and training principles of collecting past and present data to describe and prescribe the behavior of future or unseen data with respect to time. This study's objective is to build an efficient time series model in forecasting flood magnitude as a natural phenomenon that occurs frequently in nature. The study employed the combination of autoregressive (AR) and moving average (MA) technique called the ARIMA's model in time series to predict the occurrence of flood with respect to time. The implementation was done using Python in ANACONDA for simulation.

Little or no work has been done in the area of exploring auto-regressive (AR) and moving average (MA) techniques in time series to forecast flood magnitude.

The performance of time series model depends mainly on the order of the data. There are three major classifications of data in time series namely: cross sectional, time series data and the combination of both. A full blown time series data usually contain the time component as well as

series of variables at a given instance of time. We are considering only the time series component. The time series data are multivariate data collected across a given period in a sequential manner and equally spaced time of interval. Time series is the collection of observations at a regular time interval for certain variables over a given period of time. This can be adopted to predict and forecast flood magnitude, sales data, rain fall, stock market analysis, work load projection, census analysis, process and quality control systems etc. There are three major components identified in time series; namely: trend, seasonal and irregular component by [5]. The trend is a long-term upward or down ward movement, seasonal deals with the inter year fluctuations repeatable over the entire length of time and irregular depicts the random movement of data patterns. The objective of time series is to extract information from historical data and is used to predict future behavior of the same data based on previous pattern. There are two major approaches adopted in time series forecasting namely; decomposition which extracts individual components of time series and regression which uses the concept of regression on past data.

1.2 Statement of the problem

Several countries around the globe are experiencing heavy down pour (rainfall) and over flowing river banks leading to an unexpected flood disaster. This can negatively affect some part or entire localities in our living areas and frequently occurs as a natural disaster resulting into loss of life, destruction of properties and the environment at large. It exposes a number of people to be at risk of becoming more vulnerable to flood disaster. Little or no work has been done in the area of exploring auto-regressive (AR) and moving average (MA) techniques in time series to forecast flood magnitude. The aim is to build an efficient time series model in forecasting flood magnitude as a natural phenomenon that occurs frequently in nature. We are employing the combination of autoregressive (AR) and moving average (MA) technique called the ARIMA's model in time series to predict the occurrence of flood and its magnitude with respect to time.

II REVIEW OF RELATED LITERATURE

Several flood prediction schemes have been presented in the literature. Research has also shown that due to advancement in technology, flood prediction methods have moved on from solely mathematical modeling or physical parameter based flood forecasting schemes, to methodologies focused around algorithmic approaches. Flood Prediction Models (FPM) predicts flood relying on several factors such as amount of rainfall, temperature, river water level, degree of soil permeability, degree of ground saturation etc.The variables often used in these studies are

rainfall data, river water run-offs, temperature, precipitation etc. The tools used in predicting the flood level or magnitude ranges from rainfall, temperature run-off models, hydraulic models, data driven hydrological models [6].

Reference [7] developed a model that attempted forecasting the magnitude of flood in Afikpo North East of Nigeria using Artificial Neural Network and a modified runoff equation. Temperature and rainfall data were the main variables they used; their model produced 84.7% and 95.5% level of performance in rainfall and temperature respectively. However, the result cannot be relied on owing to the size of the dataset used. The dataset used in training the model was very small; temperature and rainfall data, 2012 to 2018. Also, the temperature should have been recorded at various time intervals of each day to allow for a better performance of the model which is a time-dependent event. Reference [8] identified the presence of tangible differences in the course of estimating flood magnitude of a region. They used Annual Maximum Flood (AMF) time series to determine the trend at local and regional scale using time-dependent conditions while ensuring that the factors that lead to variations in flood in an area or region remain constant for a specified period. Monte Carlo experiments were used to evaluate the performance of the model. However, the factors that were considered in their study that brought non-stationarity may change from one country to another. Reference [9] considered a situation where the physical characteristics of drainage basins could be applied in flood frequency analysis in Scotland. They used the Normal Mixtures (NORMIX) multivariate clustering algorithm. Their method was applied to about 168 basins in Scotland and five regions were identified and produced a homogeneous result of flood frequency that corresponded to the basin types. But this method, no matter how encouraging the results are, would have lapses when implemented in a third-world country like Nigeria due to topography and other development indices.

Reference [10] looked at the prediction of rainfall-based river flood forecasting in Malaysia using Nonlinear Autoregressive Exogenous model (NARX). The use of Bayesian regularisation for training of algorithm put up a better performance when compared with Levenberg-Marquardt back propagation training technique. NARX in rainfall based flood prediction showed a 99% performance. However, this was just a prediction of 24 hours in advance. In the flood magnitude prediction system proposed in this article, the prediction is targeted at months and even years ahead.

Reference [11] developed a five-hour flooding model for Kuala Lumpur applying Neural System Autoregressive Model with Exogenous feedback (NNARX). The model was implemented using

Matlab Neural Network Toolbox. The training was done with 131 samples, validation with 84 samples, and testing vector was done with 170 sample size. The training was done with gradient descent algorithm. The model was fitted at 89.2% performance level. But, the data used was small to be used to predict flood on monthly or even on yearly basis, as such the likely reason for the poor performance seen.

Reference [12] sought to use NARX to do a five-hour in advance flood prediction model for Terengganu River. 641 data sample was used for training; another 641 data sample was used for validation while 1351 was used for testing the model. The Root Mean Square Error (RMSE) value was 0.220, and the performance level of the model recorded above 80% as the model predicted the flood occurrence in advance of five hours. But the non-linearity of the data appeared to have affected the performance of the model. Again, data scaling and the number of data samples used affected the performance.

Reference [13] used NARX NN for the prediction of the dynamic characteristics of water flood level. Same study used Extended Kalman Filter (EKF) algorithm in comparison with NARX NN. NARX outperformed EKF algorithm with RMSE of 0.06m compared with 0.67m respectively. Reference [14] sought to use NARX to do a 10-hour in advance prediction of flood with the Kelang River as a case study. Results showed that NARX design was able to do the prediction 10.833 hours before time. The performance level was quite impressive with RMSE of 0.0806m. But, as stated hitherto, this case is specific to a particular river; not a region. Again, the variables like temperature, runoffs, etc. were not put into the mix.

Reference [15] used Support Vector Machine (SVM) technique to investigate a two-year knowledge based on 2005 and 2006. A total of 7 lake floods in downtown of Chiang Mai, Thailand were used as data. The study revealed that SVM performed better than Multilayer Perceptron Models (MLP) models, and that SVM can be effective in forecasting of flooding. Also the model can be used in real-world flood prediction systems with real life available data. Reference [16] also used SVM to forecast flooding event. Bird Creek,U.S.A catchment area data was used for training; and the unseen data for testing and model implementation. Rainfall and river flow data were collected from 12 rain gauges kept in the catchment area. A comparison with other prediction algorithm was made after implementation. SVM did better than all the other algorithms in the time data series. The result showed that linear and nonlinear kernel characteristics could provide significant activities against each other for various conditions in the same catchment. Reference [17] aimed at determining better attribute selection, and discharge

level estimation using Rawal Lake, Islamabad as case study. SVM, RBF (Radial-Basis-Function), ANN (Artificial Neural Network) and ARIMA were used to predict the discharge time. The major pitfall of their model is prediction over small duration.

III. MATERIALS AND METHODS

In this paper, we are focusing on the use of autoregressive integrated moving average (ARIMA), autoregressive (AR) and moving average (MA) processes. The ARIMA's model or mixed process takes into account past and present data and decomposes it through an autoregressive (AR) process that stabilizes or makes the data stationary. The moving average (MA) processes can easily carryout forecasting with or without previous error terms [18].

Dataset: The dataset used was obtained from the kaggle online repository site [19,] containing one hundred and one (101) items with attributes of date, level and target. The proposed system dataset is divided into 80% training and 20% testing set in detecting flood levels.

Table 1: Flood dataset

	YEAR	FLOOD_LEVEL	TARGET
0	1920-12-01	3	High
1	1921-12-01	1	Low
2	1922-12-01	2	Normal
3	1923-12=01	3	High
4	1924-12-01	1	Low
---	---	---	---
96	2016-12-01	2	Normal
97	2017-12-01	2	Normal
98	2018-12-01	1	Low
99	2019-12-01	2	Normal
100	2020-12-01	8	High

The autoregressive integrated moving average model (ARIMA): The ARIMA model is a more complex technique made up of the autoregressive represented with AR (p) and moving-average with MA (q). The ARIMA (p,q) model is the linear combination of the actual past values and errors terms [5]. Where q is the order of moving average and p is the order of the autoregressive component and the general form is given as:

$$Y_i = \mu + \alpha_1 Y_{t-1} + \alpha_2 Y_{t-2} + \ldots + \alpha_p Y_{t-p} + e_i - \theta_1 e_{i-1} - \ldots - \theta_q e_{i-q} \quad (1)$$

$$Y_i = \sum_{k=1}^{p} \alpha_k Y_{t-k} - \sum_{k=1}^{q} \theta_k e_{t-k} + \mu + e_t \quad (2)$$

The ARIMA model has an autocorrelation that decreases as the residual distance increases

The general form of the Autoregressive integrated Moving Average (ARIMA) is given as

$$Z_t = \sum_{k=1}^{p} \alpha_k Z_{t-k} - \sum_{k=1}^{q} \theta_k Z_{t-k} + \mu + e_t \quad (3)$$

These models were simulated using the Python Spider IDE. In this environment, the "statsmodels" library was invoked and used in training the ARIMA model and created the model outcome using the steps stated below.

The AR model: The autoregressive(AR) techniques is one of the most popular known method used in time series analysis where the dependent variables depend upon values of the response variable measured in the past [20]. It represents a linear regression on itself in the context of time series analysis [21].

$$Y_t = \beta_0 + \beta_1 * Y_{t-1} + Error_t \quad (4)$$

The AR model of the first order is represented with AR (1) model and the time lag is taken to be yearly. The Y_{t-1} is used to represent the value of Y of the previous flood year while $Error_t$ as the standard error term.

Moving Average (MA): Is a time series model that the target or output depends linearly on past and current data with respect to a time lag. The MA is represented with MA(q) where q is the the order and Ɛ is the standard error.

$$X_t = \mu + \varepsilon_t + \theta_1 \varepsilon_{t-1} + \ldots + \theta_q \varepsilon_{t-1} \quad (5)$$

Where μ is the mean. $\theta_1, \theta_2, + \ldots + \theta q$ are the parameters of the MA model and $\varepsilon_t, \varepsilon_{t-1}, + \ldots \varepsilon_{t-q}$ are the error terms. The equation 5 above can be rewritten as follows:

$$X_t = \mu + (1 + \theta_1 B + \ldots + \theta_q B^q)\varepsilon_t \quad (6)$$

Where the error value is computed using:

$$Error_t = \text{Expected-Predicted} \quad (7)$$

Algorithm 1: Moving Average(MA) Time Series Model

Step	**Processes involved**
1	Start
2	Invoke necessary libraries
3	Define and call the MA model
5	Pass in p and d parameters as the order of moving average and auto-regression
6	Call the fit() function to train model using the training dataset
7	Make predictions using the predict() function using testing dataset

8	Stop

Algorithm 2: Autoregressive(AR) Time Series Model

Step	**Processes involved**
1	Start
2	Invoke necessary libraries
3	Check for stationary
5	Determine order of AR model
6	Train model using training dataset
7	Make the predictions and evaluate predictions using testing data
8	Stop

Performance evaluation

The performance evaluation step serves as the final step used to effectively measure the success rate of MA and AR models using the rolling means and standard deviation. This study employed the coefficient of determination and standard error diagnostic tools to compare the performance of both techniques. The results are presented and discussed next.

IV. RESULTS AND DISCUSSION

We discussed about the results of MA and AR models obtained from the proposed system dataset in time series. The implementation was done with some fine-tuned hyper-parameter values for better performance. The fine tuning was achieved through the seasonal graph plot and by obtaining the rolling mean and standard deviation of the input data. The essence of fine tuning the dataset is set standardize it, thus giving rise to a stable model.

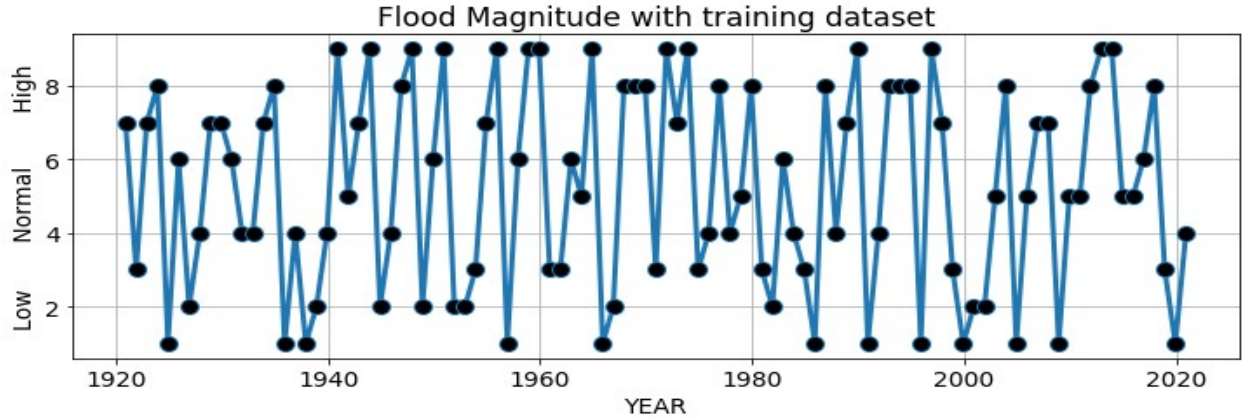

Figure1: Time series plot of flood magnitude

Figure 1 shows the up and down movement or trend of flood level that changes with previous dataset. The graph is non stationary which changes with respect to time. The graph represents the datasets (inputs) used for the prediction, that is, the data fed into the model. The time series plot of flood magnitude was implemented via the code snippet below.

```
dataset = pd.read_excel(file2, "Sheet3")
dataset['YEAR']=pd.to_datetime(dataset['YEAR'], infer_datetime_f
print(dataset)
dataset = dataset.drop('TARGET',axis=1)
indexeddataset=dataset.set_index(['YEAR'])

print(indexeddataset)

#plot graph
plt.xlabel('YEAR')
plt.ylabel('Low        Normal        High')
plt.title('Flood Magnitude with training dataset')
plt.plot(indexeddataset, lw=3, marker='o', ms=10, markerfacecolc
plt.show()

y=indexeddataset['FLOOD']
y.plot(figsize=(15, 6))
plt.show()
```

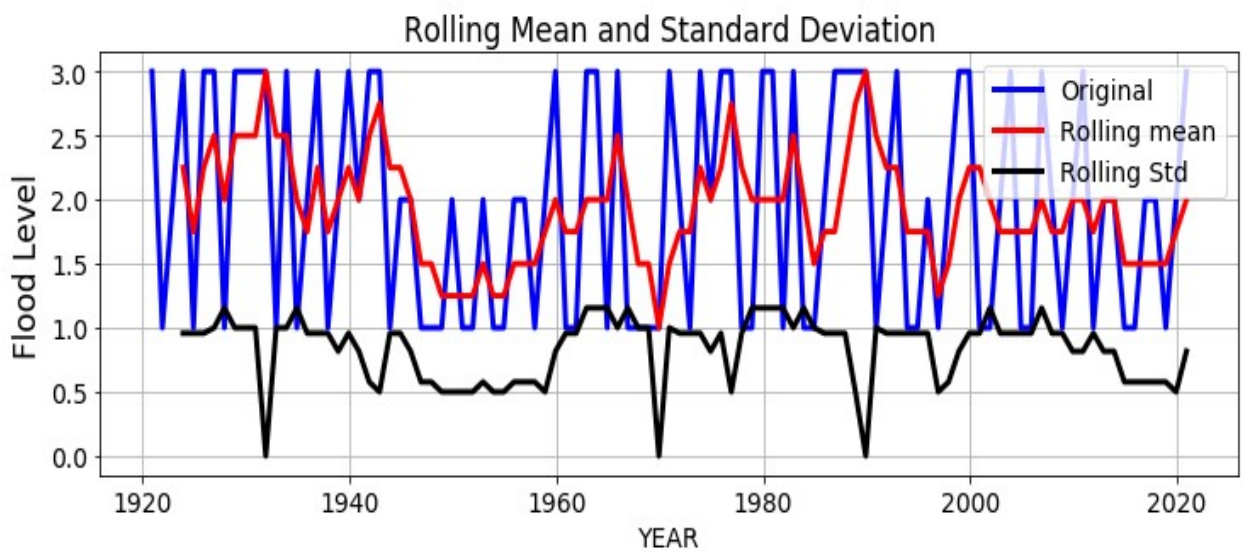

Figure 2: The rolling mean and standard deviation of flood level

Figure 2 is the rolling mean and standard deviation of flood level; it changes along with time because it's non stationary. Also; the rolling mean and the standard deviation are starting from a different year (1925) at the Year-axis because they don't have the values for the first 5 years as rolling terms (mean and standard deviation). Some data patterns appear to be seasonal, some in upward (ever increasing) and some in downward (ever decreasing) pattern; the rolling mean and standard deviation were introduced to standardize the data pattern so as to stabilize the model.

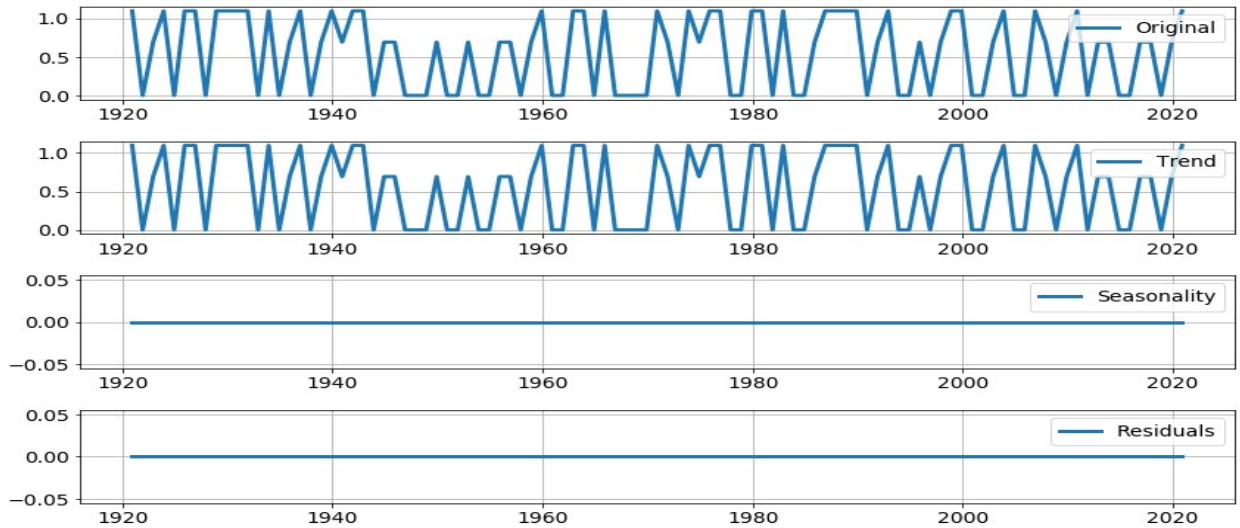

Figure 3: the seasonal decomposition graph

Figure 3 is the seasonal decomposition graph containing the original, trend and seasonal data or graph. The seasonal decomposition graph is used as a tool to make our dataset stationary. This separates the seasonal part from the trending part which then gave the last kind of graph (called residual) within the range of dataset. At the topmost part of the graph is pattern of the original data obtained, the second and third parts of the graph represents the data in its trend pattern (that is, the up and down pattern) and its seasonal (fluctuation) pattern. The last part of the graph (residual) contains the pattern of the original data after the fine tuning (separation of both seasonal and trend patterns) was completed.

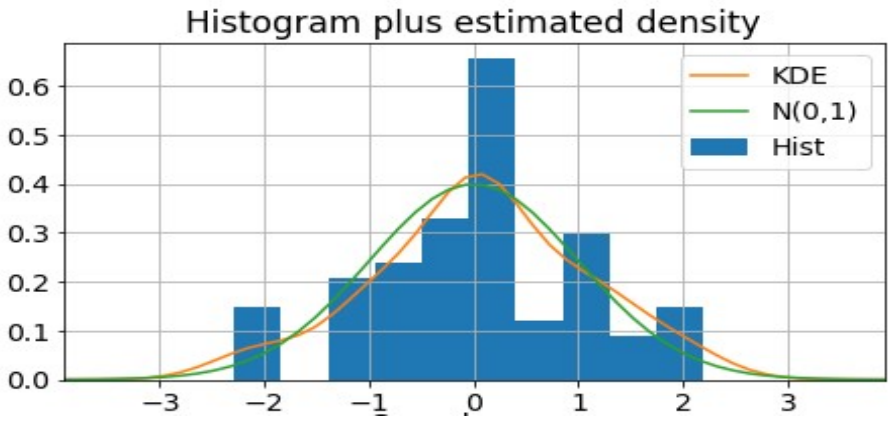

Figure 4a: Histogram plus estimated density

The curve in figure 4a shows the estimated density plot of the ARIMA's model which is essentially a smooth version of the histogram. The histogram shows a wider bandwidth which resulted in more smoothing of distributed data and even though the data was limited to fall within -3 to 3 inclusively, the density plot extended it beyond these limits. The model learns from the

data distribution and the fine-tuning or change in kernel hyper parameter value changes the distribution data drawn at each data point in forming a more smoothing curve that falls within the normal distribution curve showing the relationship. The intercepts on the curve depicts a close relationship between the independent variables and the target.

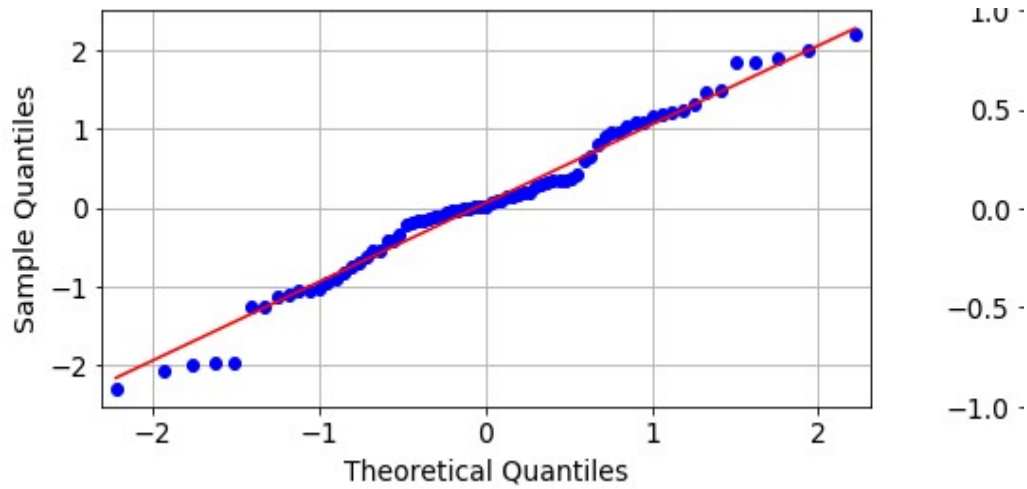

Figure 4b: The Normal quantile-quantile (Q-Q) plot

Figure 4b presents the normal quantile-quantile (Q-Q) plot. The plot is a graphical plot used to compare the probability distribution of two variables by plotting sample quantiles against theoretical quantiles. There are four outliers evident at the bottom left and three at the top right of the Normal Q-Q plot. It shows that the data points are normally and linearly distributed, thus implying that the model has good precision, less error margin and good performance.

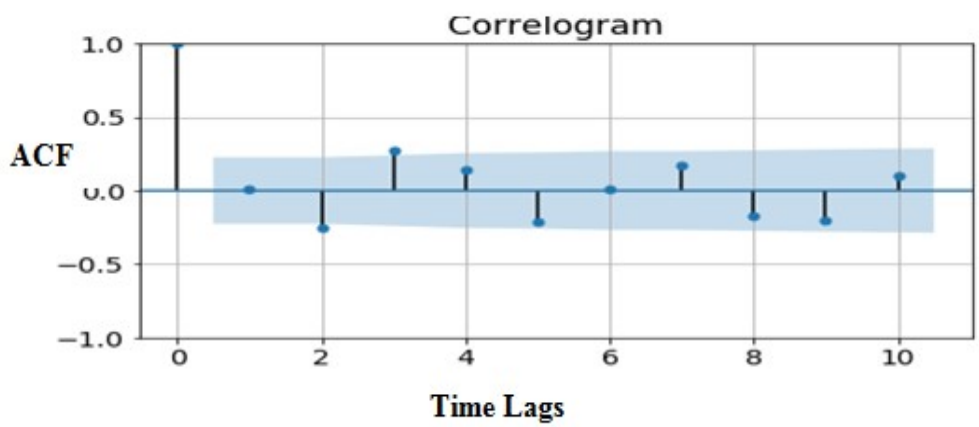

Figure 4c: The correlation graph

The correlogram in figure 4c shows the density plot which is essentially a smoothing version of the histogram and even though we limited our data plot to -2 to 3; the density plot extends beyond the limits. It is used to interpret a set of autocorrelation coefficients in which

autocorrelation function (ACF) is plotted against the time lag. We are expected to have one significant value under pure randomness in every ten lag. There are distinct spikes at lag 2 and 3; very high and equal to ACF but other values are relatively smaller in the correlogram. This also means that the model performance is within an acceptable range.

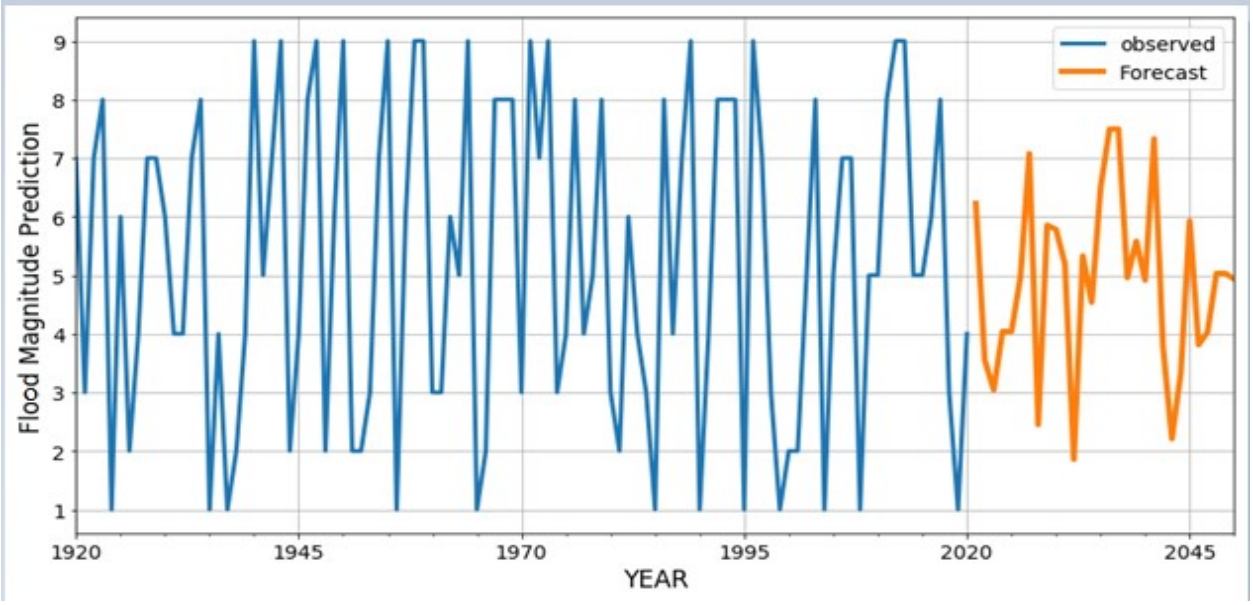

Figure 5: The prediction graph of flood level

Figure 5 is the graph showing the past and present data in predicting the future behavior of flood level (magnitude) with respect to time. The model was fed with previous data patterns to predict from the year 2022 to 2045 as shown in figure 5. The model predicted flood level of year 2022 as normal, year 2023-high, 2024-low, 2025-normal, 2026-high etc. The code snippet below was used for the actual prediction.

```
pred_uc = results.get_forecast(steps=30)
pred_ci = pred_uc.conf_int()
ax = y.plot(label='observed', figsize=(14, 7), lw=3)
pred_uc.predicted_mean.plot(ax=ax, label='Forecast', lw=4)
ax.fill_between(pred_ci.index,
                pred_ci.iloc[:, 0],
                pred_ci.iloc[:, 1], color='k', alpha=.25)

ax.set_xlabel('YEAR', fontsize=17)
ax.set_ylabel('Flood Magnitude Prediction', fontsize=17)
plt.legend()
plt.show()
```

Model Evaluation

Table I: Diagnostic Summary of the Proposed Model

	coef	Std err	Z	P>|Z|	[0.025	0.975]

ar.L1	0.0554	0.170	0.325	0.745	-0.279	0.389
ma.L1	-0.9077	0.065	-14.039	0.00	-1.034	-.0781

Table I shows the attributes of dependent and predictor variables (ar.L1, and ma.L1) with their error margin. The model Coefficient of determination is the value used in predicting the target from the independent variable. The coefficient of AR technique produced 0.0554 which is higher and performed better than the MA model that gave value as -0.9077. The higher the coefficient of determination value the better and more successful the models' performance. The standard error value of MA model gave 0.065 and AR produced 0.170 which was used to determine the significance of different parameters at lag one (L1) within the interval of 0.025 and 0.975. The MA model produced lesser error value in prediction compared to the AR model.

IV. CONCLUSION

Flood magnitude prediction model can drastically reduce the adverse effects associated with flood disaster. This will help the concerned organizations to make decisions and plan ahead as a mitigation measure. The residents within flood prone areas can be alerted on time so as to evacuate their properties on time.

ARIMA time series method of flood magnitude prediction has not been fully utilized. The developed model can serve as benchmark to other researchers since the model is scalable; little modifications can be made to predict other natural disasters such as drought by varying the determinant parameter(s). The result depicts that the model performed well, has small error margin and high degree of precision.

The use of Python in the simulation made the task less tedious. This is because Python have inbuilt libraries, functions and tools that we utilized both for simulation, prediction and diagnostic summary of the model performance. All these were achieved with few lines of codes, thus code optimization was utilized at its best.

REFERENCES

[1] Milli, L., Yacine, C., and Patrick, P. (2011). "Electric Load Forecasting Based on Statistical Robust Methods". IEEE Transactions on Power Systems, 2011. Available https://www.researchgate.net/.../224185786.

[2] Nkoana, R. (2011). " Artificial Neural Network Modeling Of Flood Prediction And Early Warning". Faculty of Natural And Agricultural Science, University of the Free State Bloemfontein.

[3] Raihan, U. I (2017). "Wireless Sensor Network Based Flood Prediction Using Belief Rule Based Expert System". Mobile and Pervasive Computing, Lulea University of Technology.

[4] Otu, O. A., Osuji, G. A., Jude, O., Ifeyinwa, M. J. and Andrew, I. L. (2013). "Application of SARIMA Models in Modelling and Forecasting Nigeria's Inflation Rates," *American Journal of Applied Mathematics and Statistics(AJAMS), vol.2, Issue.1, 16-28.*

[5] Hamidreza, M. and Leila, S. (2012). "Using SARFIMA model to Study and Predict the Iran's Oil Supply," *International Journal of Energy Economics and Policy(IJEEP), vol. 2, Issue. 1, 41-49.*

[6] Sankhua R. N. and Srivastava A. K. (2006). "Flood forecasting techniques," National Water Academy, CWC, Pune. Rretrieved from: http://ftp.comet.ucar.edu/memory-stick/flood_forecasting_techniques/index.htm#FFT.

[7] Adannaya I. S., Edim E.A., Eteng I.E., Ukegbu C.C. (2021). "Design of a flood magnitude prediction model using algorithmic and mathematical approaches". International Journal of Information Technology, 13(4), 1569 - 1579.

[8] Cunderlik J.M. and Burn D.H. (2004). "Non-stationary pooled flood frequency analysis". Journal of Hydrology vol. 276, Issue. (1-4), 210-223.

[9] Acreman M.C., Sinclair C.D. (2003). "Classification of drainage basins according to their physical characteristics; an application for flood frequency analysis in Scotland". Journal of Hydrology, Vol. 84, Issues. 3–4, 365-380.

[10] Noor H.M, Ndzi D., Yang G.and Safar N.Z.M. (2017). "Rainfall-based River Flow Prediction Using NARX in Malaysia", IEEE 13th International Colloquium on Signal Processing & Its Applications (CSPA), BatuFerringhi, 67-72.

[11] Adnan R., Samad A.M., Zain Z.M. and Ruslan F.A. (2014). "5 hours flood prediction modelling using improved NNARX structure: case study Kuala Lumpur". The IEEE 4th International Conference on System Engineering and Technology (ICSET), Bandung, 1-5.

[12] Lukman Q.A., Ruslan F.A. and Adnan R. (2014). "5 Hours ahead of time flood water level prediction modelling using NNARX Technique: Case Study Terengganu," The 7th IEEE Control and System Graduate Research Colloquium (ICSGRC), Shah Alam, 104-108.

[13] Ruslan F. A., Samad A.M., Zain Z.M. and Adnan R. (2013). "Flood prediction using NARX neural network and EKF prediction technique: A comparative study," 2013 IEEE 3rd International Conference on System Engineering and Technology, Shah Alam, 203-208.

[14] Ruslan F. A., Samad A.M., Zain Z.M. and Adnan R. (2014). "Flood water level modeling and prediction using NARX Neural Network: Case Study at Kelang River," IEEE 10th International Colloquium on Signal Processing and its Applications, Kuala Lumpur, 204-207.

[15] Theera-Umpon N., Auephanwiriyakul S., Suteepohnwiroj S., Pahasha J. and Wantanajittikul K. (2008). "River Basin Flood Prediction using Support Vector Machines," The IEEE International Joint Conference on Neural Networks(IEEE World Congress on Computational Intelligence), Hong Kong, 3039-3043.

[16] Han D., Chan L., and Zhu N. (2007). “Flood forecasting using support vector machines”., Journal of hydroinformatics, 9(4), 267-276.

[17] Ali M., Qamar A. M., and Ali B. (2013). “Data analysis, discharge classifications, and predictions of hydrological parameters for the management of Rawal Dam in Pakistan”, *Proceedings of 12th International Conference on Machine Learning and Applications*.1, 382-385.

[18] Fritzer, F., Gabriel, M. and Johann, S.(2002). "Forecasting Austrian HICP and its Components Using VAR and ARIMA Model," Working Paper 73, Oesterreichische National Bank(Austrian Central Bank), 73.

[19] Kaggle Online (2021). https://www.kaggle.com/ziweritinstanley/flood-level-dataset.

[20] Iwok, I. (2016). Method of Estimating Missing Values in a Stationary Autoregressive (AR) Process, Asian Journal of Mathematics & Statistics, 9(1), 6-10.

[21] Maleika, W. (2014). Moving Average Optimization in Digital Terrain Model Generation Based on test Multibeam Echosounder data, Geo-Marine Letters, 35(1), 61-68.

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Adaptive, autonomic and context-aware computing	**Location Based Services**
Advanced Computing Architectures and New Programming Models	**Management information systems**
Agent-based middleware	**Medical imaging**
Autonomic and self-managing middleware	**Micro/nano technology**
B2B and B2C management	**Middleware Issues**
BioInformatics	**Middleware services and agent technologies**
Bio-Medicine	**Mobile and Wireless Networks**
Biotechnology	**Mobile Computing and Applications**
Broadband and intelligent networks	**Mobile networks and services**
Broadband wireless technologies	**Multimedia Communications**
Cloud Computing and Applications	**Multimodal sensing and context for pervasive applications**
Collaborative applications	**Multisensor fusion**
Communication architectures for pervasive computing	**Natural Language Processing**
Communication systems	**Network management and services**
Computational intelligence	**Network Modeling and Simulation**
Computer and microprocessor-based control	**Network Performance; Protocols; Sensors**
Computer Architecture and Embedded Systems	**Networking theory and technologies**
Computer Business	**Neural Networks**
Computer Vision	**Neuro-Fuzzy and applications**
Computer-based information systems in health care	**Open Models and Architectures**
Computing Ethics	**Open Source Tools**
Context-awareness and middleware	**Operations research**
Cross-layer design and Physical layer based issue	**Optical Networks**
Cryptography	**Pattern Recognition**
Data Base Management	**Peer to Peer and Overlay Networks**
Data Mining	**Perception and semantic interpretation**
Data Retrieval	**Pervasive Computing**
Decision making	**Performance optimization**
Digital Economy and Digital Divide	**Positioning and tracking technologies**
Digital signal processing theory	**Programming paradigms for pervasive systems**
Distributed Sensor Networks	**Quality of Service and Quality of Experience**
E-Business	**Real-time computer control**
E-Commerce	**Real-time information systems**
E-Government	**Real-time multimedia signal processing**
Emerging signal processing areas	**Reconfigurable, adaptable, and reflective middleware approaches**
Enabling technologies for pervasive systems (e.g., wireless BAN, PAN)	**Remote Sensing**
Encryption	**RFID and sensor network applications**
Energy-efficient and green pervasive computing	**Scalability of middleware**
Event-based, publish/subscribe, and message-oriented middleware	**Security and risk management**
Evolutionary computing and intelligent systems	**Security middleware**

Expert approaches **Fuzzy algorithms** **Fuzzy logics** **GPS and location-based applications** **Green Computing** **Grid Networking** **Healthcare Management Information Technology** **Human Computer Interaction (HCI)** **Image analysis and processing** **Image and multidimensional signal processing** **Image and Multimedia applications** **Industrial applications of neural networks** **Information and data security** **Information indexing and retrieval** **Information Management** **Information processing** **Information systems and applications** **Information Technology and their application** **Instrumentation electronics** **Intelligent Control System** **Intelligent sensors and actuators** **Internet applications and performances** **Internet Services and Applications** **Internet Technologies, Infrastructure, Services & Applications** **Interworking architecture and interoperability**	**Security, Privacy and Trust** **Security Systems and Technolgies** **Sensor array and multi-channel processing** **Sensor fusion** **Sensors and RFID in pervasive systems** **Service oriented middleware** **Signal Control System** **Signal processing** **Smart devices and intelligent environments** **Smart home applications** **Social Networks and Online Communities** **Software Engineering** **Software engineering techniques for middleware** **Speech interface; Speech processing** **Supply Chain Management** **System security and security technologies** **Technology in Education** **Theoretical Computer Science** **Transportation information** **Trust, security and privacy issues in pervasive systems** **Ubiquitous and pervasive applications** **Ubiquitous Networks** **User interfaces and interaction models** **Virtual reality** **Vision-based applications** **Web Technologies** **Wired/Wireless Sensor** **Wireless technology**